Demonstration of classification rule process on dataset employee.arff using naïve bayes algorithm.

Aim: This experiment illustrates the use of naïve bayes classifier in weka. The sample data set used in this experiment is "employee"data available at arff format. This document assumes that appropriate data pre processing has been performed.
Steps involved in this experiment:
1. We begin the experiment by loading the data (employee.arff) into weka.
Step2: next we select the "classify" tab and click "choose" button to select the "id3"classifier.
Step3: now we specify the various parameters. These can be specified by clicking in the text box to the right of the chose button. In this example, we accept the default values his default version does perform some pruning but does not perform error pruning.
Step4: under the "text "options in the main panel. We select the 10-fold cross validation as our evaluation approach. Since we don't have separate evaluation data set, this is necessary to get a reasonable idea of accuracy of generated model.
Step-5: we now click"start"to generate the model .the ASCII version of the tree as well as evaluation statistic will appear in the right panel when the model construction is complete.
Step-6: note that the classification accuracy of model is about 69%.this indicates that we may find more work. (Either in preprocessing or in selecting current parameters for the classification)
Step-7: now weka also lets us a view a graphical version of the classification tree. This can be done by right clicking the last result set and selecting "visualize tree" from the pop-up menu.
Step-8: we will use our model to classify the new instances.
Step-9: In the main panel under "text "options click the "supplied test set" radio button and then click the "set" button. This will show pop-up window which will allow you to open the file containing test instances.

Description:
Bayesian classification is based on Baye's Theorem. Bayesian classifiers are the statistical classifiers. Bayesian classifiers can predict class membership probabilities such as the probability that a given tuple belongs to a particular class.

Baye's Theorem

Bayes' Theorem is named after Thomas Bayes. There are two types of probabilities

- Posterior Probability [P(H/X)]
- Prior Probability [P(H)]

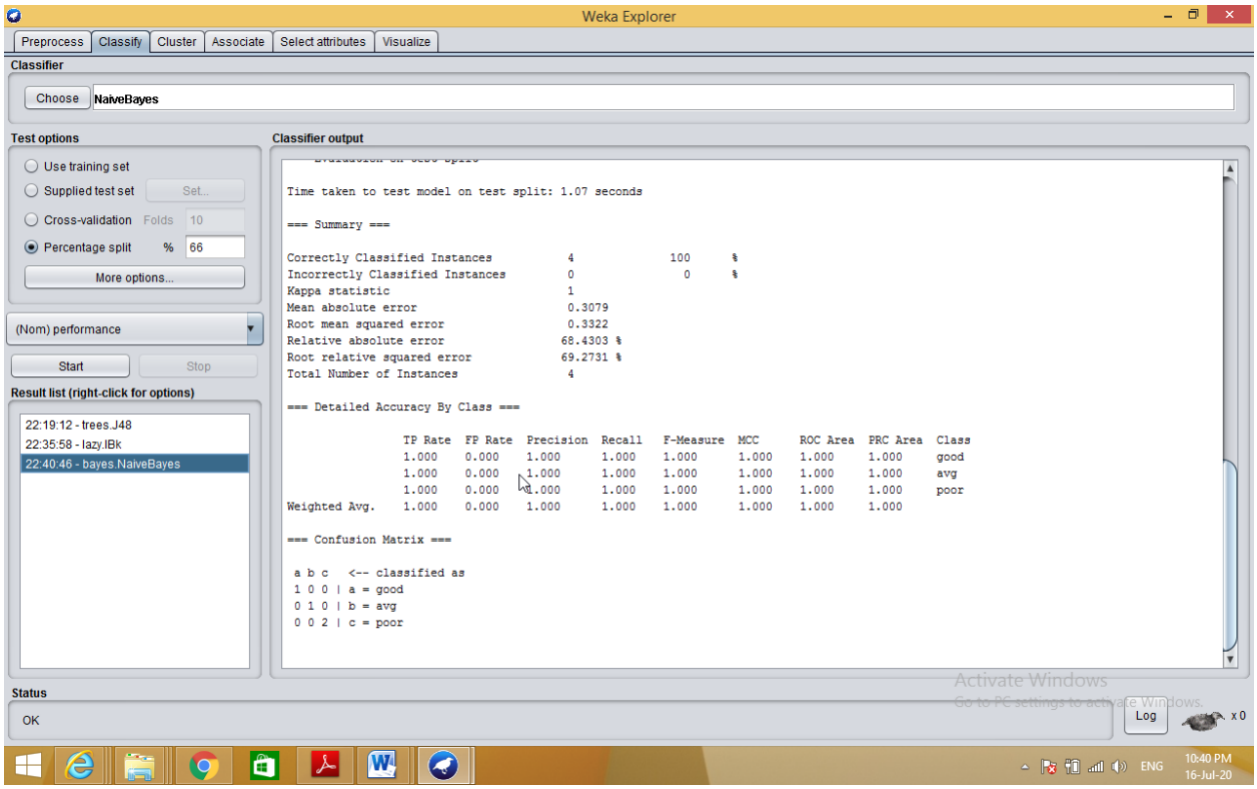
where X is data tuple and H is some hypothesis.
According to Bayes' Theorem,

P(H/X)= P(X/H)P(H) / P(X)

Procedure:

Step 1: Load appropriate dataset into WEKA

Step 2: Go to Classify tab→Bayes→NaiveBayes



Step 3: choose ->naive bayes ; Test options ->supplied test set ->open file

Step 4: more options :output predictions ->output test predictions on test set

Data set : **classification rule process on dataset employee.arff using naïve bayes algorithm.**

@relation employee

@attribute age {25,27,28,29,30,35,48,39,45}

@attribute salary {10k,15k,17k,20k,25k,30k,34k,35k,32k,33k,40k}

@attribute performance {good,avg,poor}

@data

%

25,10k,poor

27,15k,poor

27,17k,poor

28,17k,poor

29,20k,avg

30,25k,avg

29,20k,avg

30,20k,avg

35,32k,good

39,33k,good

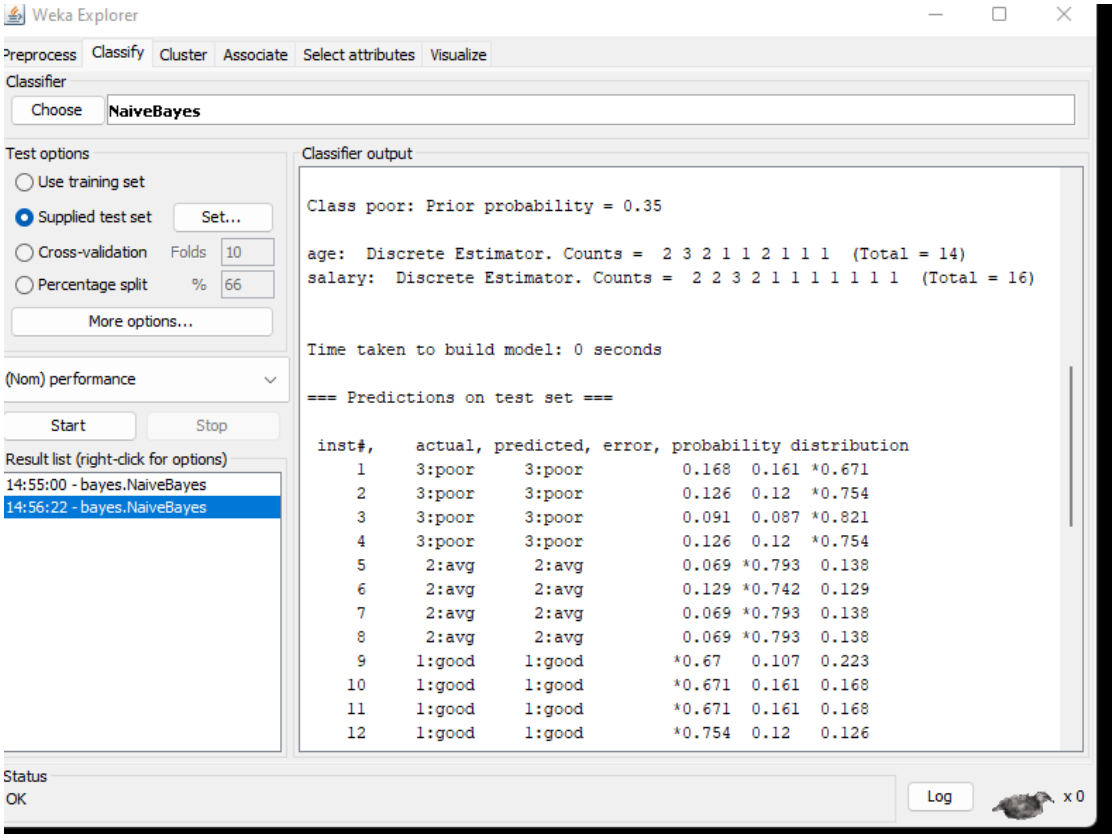
45,40k,good

48,34k,good

48,32k,good

35,20k,poor

%


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
NaiveBayes
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) performance
Start
Stop
Result list (right-click for options)
14:55:00 - bayes.NaiveBayes
14:56:22 - bayes.NaiveBayes
Classifier output
Class poor: Prior probability = 0.35
age: Discrete Estimator. Counts = 2 3 2 1 1 2 1 1 1 (Total = 14)
salary: Discrete Estimator. Counts = 2 2 3 2 1 1 1 1 1 1 1 (Total = 16)
Time taken to build model: 0 seconds
=== Predictions on test set ===
inst#, actual, predicted, error, probability distribution
1 3:poor 3:poor 0.168 0.161 *0.671
2 3:poor 3:poor 0.126 0.12 *0.754
3 3:poor 3:poor 0.091 0.087 *0.821
4 3:poor 3:poor 0.126 0.12 *0.754
5 2:avg 2:avg 0.069 *0.793 0.138
6 2:avg 2:avg 0.129 *0.742 0.129
7 2:avg 2:avg 0.069 *0.793 0.138
8 2:avg 2:avg 0.069 *0.793 0.138
9 1:good 1:good *0.67 0.107 0.223
10 1:good 1:good *0.671 0.161 0.168
11 1:good 1:good *0.671 0.161 0.168
12 1:good 1:good *0.754 0.12 0.126
Status
OK
Log
x 0